

Data & AI Case Study

Mapped to two Wuerth internship roles

DISCLAIMER: Independent analysis based on PUBLIC information only. NOT affiliated with, endorsed by, or reviewed by the Wuerth Group. No internal, confidential, or proprietary Wuerth data or systems were used. Wuerth-scale figures are public and approximate. All portfolio performance numbers are measured on SYNTHETIC / self-generated data and demonstrate method only -- they are not claims about Wuerth's real business.

This is my independent case study. I mapped the requirements of two Wuerth Data & AI internship postings to work I have already built and measured in my own portfolio. The euro and accuracy figures throughout are on synthetic data and exist to prove the methods work and are measurable -- not to forecast Wuerth outcomes.

Job #1 -- (Agentic) Automation with Low-code Platforms

Agentic AI workflows, low-code (n8n / Power Automate), connecting systems & APIs, document automation, ROI, rapid prototyping.

Job #2 -- Data & AI Analytics

BI / Power BI, KPI dashboards, forecasting & predictive analytics, Python & SQL, data modelling, turning data into decisions.

Who Wuerth is (public info)

Wuerth (public profile, approximate): a large family-owned German-HQ group in assembly/fastening and industrial MRO distribution -- ~400+ companies in 80+ countries, ~87,000 employees, ~EUR 20B+ revenue, a catalogue in the millions of articles, multi-channel (field sales, branches, e-commerce, e-procurement/EDI, ORSY inventory systems). A business that runs on assortment, pricing, availability, logistics, and high-volume transactional documents -- a natural fit for applied Data & AI.

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Opportunity map (summary)

Wuerth business area (public-info) -> Data/AI approach -> repo + measured result (synthetic)

Area (public-info)	Approach	Repo	Result (synthetic)
Procurement / assortment	MILP vs greedy	revops-optimizer	MILP > greedy; ~EUR160k/yr
Pricing & margin	Elasticity + leakage	sales-kpi-analytics	EUR2.6M leakage lever
Inventory / replenishment	Newsvendor + forecast	distributor-intel-platform	MASE 0.376 (< naive)
Sales KPIs & analytics	KPIs + rolling-origin CV	sales-kpi-analytics	MASE < 1; exec PDF
Logistics / routing	OR-Tools CP-SAT VRP	route-optimizer	4.6% / 31% savings
E-procurement automation	Agentic + n8n low-code	agentic-automation-lab	~EUR625k/yr modelled
Document processing	Agentic RFQ/invoice extract	doc-extract-agent	~EUR145k/yr modelled
Customer retention	Decline classifier	revops-optimizer	ROC-AUC 0.99

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Job #1 -- (Agentic) Automation with Low-code

Every posting requirement -> repo + artifact + measured proof (all synthetic data).

Requirement	Repo / artifact	Proof (synthetic)
Build AI-agent workflows	agentic-automation-lab	agentic tool-use loops
Low-code (n8n / Power Automate)	agentic-automation-lab: n8n/rfq_intake_agent.json	RFQ-intake agent; ~EUR625k/yr
Visual / flow-based building	agent-flow-studio	flow builder; ~EUR47k/yr
Connecting systems / APIs	agentic-automation-lab connector layer	agents call external APIs
Process automation (back-office)	doc-extract-agent	RFQ/invoice -> structured
Document intake / understanding	doc-extract-agent	extract + validate; ~EUR145k/yr
Rapid prototyping	agent-flow-studio + agentic-automation-lab	runnable prototypes
Show automation ROI	automation-roi-explorer	EUR383k/yr net modelled
Python engineering	all automation repos	Python throughout
Stakeholder communication	automation-roi-explorer	business-readable ROI views

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Job #2 -- Data & AI Analytics

Every posting requirement -> repo + artifact + measured proof (all synthetic data).

Requirement	Repo / artifact	Proof (synthetic)
BI / Power BI	revops-optimizer: powerbi/DAX_measures.md	DAX pack + 3-page report
KPI dashboards / metrics	sales-kpi-analytics	KPI layer + exec PDF
Predictive analytics / forecasting	sales-kpi-analytics: rolling-origin CV	MASE < 1 vs seasonal-naive
Forecasting rigour	distributor-intel-platform	MASE 0.376
Python for analytics	sales-kpi-analytics, revops-optimizer	end-to-end pipelines
SQL / data modelling	sales-kpi-analytics SQL set	spend analysis queries
Excel-level tabular analysis	sales-kpi-analytics	spend breakdowns
Optimization / prescriptive	revops-optimizer MILP + newsvendor	~EUR160k/yr uplift
Elasticity / statistical modelling	revops-optimizer predict layer	elasticity; ROC-AUC 0.99
Data into business decisions	sales-kpi-analytics	EUR2.6M leakage lever
Logistics / ops analytics	route-optimizer	4.6% / 31% savings
Present to decision-makers	revops-optimizer exec deck	exec narrative deck